

Molecular Mechanisms and the Role of Saturated Fatty Acids in the Progression of Non-Alcoholic Fatty Liver Disease

The steady rise in Western obesity rates has been closely linked to significant increases in a multitude of accompanying health problems including Non-Alcoholic Fatty Liver Disease (NAFLD). NAFLD severity ranges from simple steatosis to acute steatohepatitis, but the molecular mechanisms controlling progression of this disease are poorly understood. Recent literature suggests that elevated free fatty acids (FFAs), especially saturated FFAs, may play an important role in lipotoxic mechanisms, both in experimental models and in NAFLD patients. This review highlights important cellular pathways involved in hepatic lipotoxicity and how the degree of intrahepatic lipid saturation controls cell fate in response to an elevated FFA load. Relevant cellular processes that have been causally linked to lipid-induced apoptosis, known as lipoapoptosis, include endoplasmic reticulum (ER) stress, oxidative stress, mitochondrial dysfunction, and Jun N-terminal kinase (JNK) signaling. In contrast, increased triglyceride synthesis has been shown to have a protective effect against lipotoxicity, despite being one of the hallmark traits of NAFLD. Developing a more nuanced understanding of the molecular mechanisms underlying NAFLD progression will lead to more targeted and effective therapeutics for this increasingly prevalent disease, which to date has no proven pharmacologic treatment to prevent or reverse its course.